Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

Expanding Brackets

Name _____ Date _____ No calculator Version 1

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KEY WORDS expand *(whakaroha)* · brackets *(taiapa)* · term *(tūnga)* · like terms *(tūnga ōrite)*

A A positive number outside ●○○

Remember: Multiply every term inside the bracket by the number outside: $3(x+4) = 3x+12$.

1. Expand.

a) $2(x+3)$ = _____ **b)** $3(x+5)$ = _____ **c)** $5(x-2)$ = _____ **d)** $4(2x+1)$ = _____

e) $6(x-4)$ = _____ **f)** $2(3x+5)$ = _____ **g)** $7(2x-3)$ = _____ **h)** $3(4x+1)$ = _____

i) $10(x+6)$ = _____ **j)** $8(3x-2)$ = _____ **k)** $9(x-9)$ = _____ **l)** $5(5x+4)$ = _____

m) $12(x+2)$ = _____ **n)** $3(7x-4)$ = _____ **o)** $20(x-5)$ = _____ **p)** $6(4x+9)$ = _____

B A negative number outside ●●○

Watch out: A negative number outside changes the sign of every term: $-2(x-5) = -2x+10$.

2. Expand.

a) $-2(x+3)$ = _____ **b)** $-3(x-4)$ = _____ **c)** $-5(2x+1)$ = _____ **d)** $-(x+6)$ = _____

e) $-4(3x-2)$ = _____ **f)** $-6(x-1)$ = _____ **g)** $-2(5x+4)$ = _____ **h)** $-(2x-7)$ = _____

i) $-7(x+2)$ = _____ **j)** $-3(4x-5)$ = _____ **k)** $-10(x-3)$ = _____ **l)** $-8(2x+1)$ = _____

m) $-(5x-9)$ = _____ **n)** $-6(2x+7)$ = _____ **o)** $-9(x-1)$ = _____ **p)** $-2(8x-3)$ = _____

C A letter outside ●●○

Remember: Remember that $x \times x = x^2$, so $x(x+3) = x^2+3x$.

3. Expand.

a) $x(x+3)$ **b)** $x(x-5)$ **c)** $2x(x+4)$ **d)** $3x(2x+1)$

e) $x(4x-1)$ **f)** $5x(x-2)$ **g)** $-x(x+6)$ **h)** $-2x(3x+1)$

i) $4x(2x-3)$ **j)** $-3x(x-4)$ **k)** $6x(x+1)$ **l)** $2x(5x+2)$

m) $x(x+10)$ **n)** $-x(2x-9)$ **o)** $8x(x-5)$ **p)** $3x(3x+4)$

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D Two different letters ●●○

4. Expand.

a) $3(2a + b)$ **b)** $5(x - 2y)$ **c)** $a(b + c)$ **d)** $2x(x + 3y)$

e) $-4(2m - n)$ **f)** $p(3p - q)$ **g)** $3a(2b + 5)$ **h)** $6(a + 2b)$

i) $y(x - y)$ **j)** $-2a(a + 4b)$ **k)** $5x(2x + y)$ **l)** $4b(3a - 2b)$

E Expand and simplify ●●●

> **Remember:** Expand each bracket, then collect like terms.

5. Expand and simplify.

a) $2(x + 3) + 3(x + 1)$ **b)** $4(x + 2) + 2(x + 5)$ **c)** $3(2x + 1) + 2(x - 4)$

d) $5(x + 1) - 2(x + 3)$ **e)** $2(3x - 1) + 3(x + 2)$ **f)** $4(x + 3) - 3(x - 1)$

g) $6(x - 2) + (2x + 5)$ **h)** $3(x + 4) - (x + 7)$ **i)** $2(2x + 5) - 2(x + 1)$

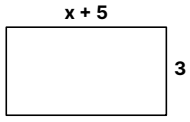
j) $5(2x + 1) + 4(x - 3)$ **k)** $7(x + 2) - 5(x + 1)$ **l)** $3(3x + 2) - 4(2x + 1)$

F Areas ●●●

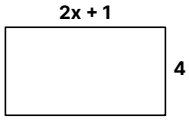
> **Remember:** Area of a rectangle = length × width. Write the area with a bracket first, then expand it.

6. Write an expression for the area of each rectangle, then expand it.

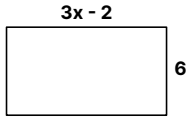
a)



b)



c)



★ Extension

7. Two brackets. Multiply every term in the first bracket by every term in the second, then simplify.

a) $(x + 2)(x + 3)$ **b)** $(x + 4)(x + 5)$ **c)** $(x + 3)(x - 2)$

d) $(x - 1)(x + 6)$ **e)** $(x - 4)(x - 3)$ **f)** $(2x + 1)(x + 3)$

g) $(x + 7)(x + 1)$ **h)** $(x + 5)(x - 5)$ **i)** $(3x + 2)(x - 1)$

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ALGEBRA

Expanding Brackets

Name _____ Date _____ No calculator Version 2

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KEY WORDS expand *(whakaroha)* · brackets *(taiapa)* · term *(tūnga)* · like terms *(tūnga ōrite)*

A A positive number outside ●○○

Remember: Multiply every term inside the bracket by the number outside: $3(x+4) = 3x+12$.

1. Expand.

a) $2(x+4)$ = _____ **b)** $3(x+6)$ = _____ **c)** $4(x-3)$ = _____ **d)** $5(2x+1)$ = _____

e) $7(x-2)$ = _____ **f)** $2(4x+3)$ = _____ **g)** $6(2x-5)$ = _____ **h)** $3(5x+1)$ = _____

i) $10(x+8)$ = _____ **j)** $9(2x-3)$ = _____ **k)** $8(x-8)$ = _____ **l)** $4(6x+5)$ = _____

m) $11(x+3)$ = _____ **n)** $3(8x-5)$ = _____ **o)** $20(x-4)$ = _____ **p)** $7(3x+8)$ = _____

B A negative number outside ●●○

Watch out: A negative number outside changes the sign of every term: $-2(x-5) = -2x+10$.

2. Expand.

a) $-2(x+5)$ = _____ **b)** $-3(x-2)$ = _____ **c)** $-4(2x+1)$ = _____ **d)** $-(x+8)$ = _____

e) $-5(3x-2)$ = _____ **f)** $-7(x-1)$ = _____ **g)** $-2(4x+3)$ = _____ **h)** $-(3x-5)$ = _____

i) $-6(x+2)$ = _____ **j)** $-3(5x-4)$ = _____ **k)** $-10(x-6)$ = _____ **l)** $-9(2x+1)$ = _____

m) $-(4x-7)$ = _____ **n)** $-5(2x+9)$ = _____ **o)** $-8(x-1)$ = _____ **p)** $-2(7x-4)$ = _____

C A letter outside ●●○

Remember: Remember that $x \times x = x^2$, so $x(x+3) = x^2+3x$.

3. Expand.

a) $x(x+4)$ **b)** $x(x-6)$ **c)** $3x(x+2)$ **d)** $2x(3x+1)$

e) $x(5x-1)$ **f)** $4x(x-3)$ **g)** $-x(x+7)$ **h)** $-3x(2x+1)$

i) $5x(2x-1)$ **j)** $-2x(x-5)$ **k)** $7x(x+1)$ **l)** $3x(4x+2)$

m) $x(x+12)$ **n)** $-x(3x-8)$ **o)** $9x(x-4)$ **p)** $2x(6x+5)$

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D Two different letters ●●○

4. Expand.

a) $2(3a + b)$ **b)** $4(x - 3y)$ **c)** $x(y + z)$ **d)** $3x(x + 2y)$

e) $-5(2m - n)$ **f)** $p(2p - q)$ **g)** $2a(3b + 4)$ **h)** $7(a + 2b)$

i) $b(a - b)$ **j)** $-3a(a + 2b)$ **k)** $4x(3x + y)$ **l)** $5b(2a - 3b)$

E Expand and simplify ●●●

> **Remember:** Expand each bracket, then collect like terms.

5. Expand and simplify.

a) $3(x + 2) + 2(x + 4)$ **b)** $2(x + 5) + 4(x + 1)$ **c)** $4(2x + 1) + 3(x - 2)$

d) $6(x + 1) - 3(x + 2)$ **e)** $3(3x - 1) + 2(x + 5)$ **f)** $5(x + 2) - 2(x - 3)$

g) $4(x - 3) + (3x + 2)$ **h)** $2(x + 6) - (x + 5)$ **i)** $3(2x + 3) - 3(x + 1)$

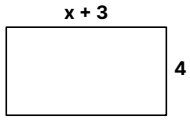
j) $4(2x + 1) + 5(x - 2)$ **k)** $8(x + 1) - 5(x + 2)$ **l)** $2(3x + 4) - 3(2x + 1)$

F Areas ●●●

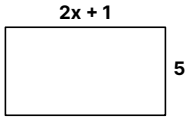
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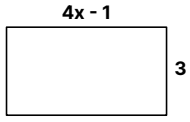
a)



b)



c)



★ Extension

7. Two brackets. Multiply every term in the first bracket by every term in the second, then simplify.

a) $(x + 3)(x + 4)$ **b)** $(x + 2)(x + 6)$ **c)** $(x + 5)(x - 3)$

d) $(x - 2)(x + 7)$ **e)** $(x - 5)(x - 2)$ **f)** $(2x + 3)(x + 1)$

g) $(x + 8)(x + 1)$ **h)** $(x + 6)(x - 6)$ **i)** $(3x + 1)(x - 2)$

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